

1. **Rate distortion function with infinite distortion.**

Find the rate distortion function $R(D) = \min_{p(\hat{x}|x): \mathbb{E}[d(X, \hat{X})] \leq D} I(X; \hat{X})$ for $X \sim \text{Bernoulli}(1/2)$ and distortion

$$d(x, \hat{x}) = \begin{cases} 0, & x = \hat{x} \\ 1, & x = 1, \hat{x} = 0 \\ \infty, & x = 0, \hat{x} = 1 \end{cases}$$

where $x, \hat{x} \in \{0, 1\}$, i.e., reconstructing a 0 by a 1 is not allowed.

Solution: Rate distortion function with infinite distortion.

We wish to evaluate the rate distortion function

$$R(D) = \min_{p(\hat{x}|x): \sum_{x, \hat{x}} p(x)p(\hat{x}|x)d(x, \hat{x}) \leq D} I(X; \hat{X})$$

Since $d(0, 1) = \infty$, we must have $p(0, 1) = 0$ for any finite distortion D . Thus, the distortion $D = p(1, 0)$, and hence we have the following joint distribution for $(X, \hat{X})$ (assuming $D \leq \frac{1}{2}$):

$$p(x, \hat{x}) = \begin{bmatrix} \frac{1}{2} & 0 \\ D & \frac{1}{2} - D \end{bmatrix}$$

The mutual information for this joint distribution is

$$\begin{aligned} R(D) &= I(X; \hat{X}) \\ &= H(X) - H(X|\hat{X}) \\ &= H(0.5, 0.5) - \left(\frac{1}{2} + D\right) H\left(\frac{1/2}{1/2 + D}, \frac{D}{1/2 + D}\right) \\ &= 1 + \frac{1}{2} \log \frac{1/2}{1/2 + D} + D \log \frac{D}{1/2 + D} \end{aligned}$$

which is the rate distortion function for this binary source if $0 \leq D \leq 1/2$. Since we can achieve $D = 1/2$ with zero rate (use $p(\hat{x} = 0) = 1$), we have $R(D) = 0$ for $D \geq 1/2$

2. **Rate distortion.**

Let $d(x, \hat{x})$ be a distortion function. We have a source $X \sim p(x)$. Let $R(D)$ be the associated rate distortion function.

- Find $\tilde{R}(D)$ in terms of $R(D)$, where $\tilde{R}(D)$ is the rate distortion function associated with the distortion $\tilde{d}(x, \hat{x}) = d(x, \hat{x}) + a$ for some constant $a > 0$. (They are not equal)
- Now suppose that $d(x, \hat{x}) \geq 0$ for all $x, \hat{x}$ and define a new distortion function $d^*(x, \hat{x}) = b \cdot d(x, \hat{x})$, where b is some number ≥ 0 . Find the associated rate distortion function $R^*(D)$ in terms of $R(D)$.

(c) Let $X \sim \mathcal{N}(0, \sigma^2)$ and $d(x, \hat{x}) = 5(x - \hat{x})^2 + 3$. What is $R(D)$?

Solution: Rate distortion.

(a)

$$\begin{aligned}\tilde{R}(D) &= \inf_{p(\hat{x}|x): \mathbb{E}[\tilde{d}(x, \hat{x})] \leq D} I(X; \hat{X}) \\ &= \inf_{p(\hat{x}|x): \mathbb{E}[\tilde{d}(x, \hat{x})] + a \leq D} I(X; \hat{X}) \\ &= \inf_{p(\hat{x}|x): \mathbb{E}[\tilde{d}(x, \hat{x})] \leq D - a} I(X; \hat{X}) \\ &= R(D - a)\end{aligned}$$

(b)

$$\begin{aligned}R^*(D) &= \inf_{p(\hat{x}|x): \mathbb{E}[d^*(x, \hat{x})] \leq D} I(X; \hat{X}) \\ &= \inf_{p(\hat{x}|x): \mathbb{E}[bd(x, \hat{x})] \leq D} I(X; \hat{X}) \\ &= \inf_{p(\hat{x}|x): \mathbb{E}[d(x, \hat{x})] \leq D/b} I(X; \hat{X}) \\ &= R(D/b)\end{aligned}$$

else if $b = 0$ then $d^* = 0$ and $R^*(D) = 0$.

(c) Let $R_{se}(D)$ be the rate distortion function associate with the distortion $d_{se}(x, \hat{x}) = (x - \hat{x})^2$. Then from previous parts, we have

$$R(D) = R_{se}\left(\frac{D - 3}{5}\right).$$

We know that

$$R_{se}(D) = \begin{cases} \frac{1}{2} \log \frac{\sigma^2}{D}, & 0 \leq D \leq \sigma^2 \\ 0, & D > \sigma^2 \end{cases}$$

Therefore, we have

$$R(D) = \begin{cases} \frac{1}{2} \log \frac{5\sigma^2}{D-3}, & 3 \leq D \leq 5\sigma^2 + 3 \\ 0, & D > 5\sigma^2 + 3 \end{cases}$$

3. Adding a column to the distortion matrix.

Let $R(D)$ be the rate distortion function for an i.i.d. process with probability mass function $p(x)$ and distortion function $d(x, \hat{x})$, $x \in \mathcal{X}$, $\hat{x} \in \hat{\mathcal{X}}$. Now suppose that we add a new reproduction symbol $\hat{x}_0$ to $\hat{\mathcal{X}}$ with associated distortion $d(x, \hat{x}_0)$, $x \in \mathcal{X}$. Does this increase or decrease $R(D)$ and why?

Solution: Adding a column

Let the new rate distortion function be denoted as $\tilde{R}(D)$, and note that we can still achieve $R(D)$ by restricting the support of $p(x, \hat{x})$, i.e., by simply ignoring the new symbol. Thus, $\tilde{R}(D) \leq R(D)$.

4. Let q be the forward process of star-diffusion model, which is given by

$$q(X^{(1)}, \dots, X^{(N-1)}, X^{(N)} | X^{(0)}) = \prod_{n=1}^N q(X^{(n)} | X^{(0)})$$

$$q(X^{(n)} | X^{(0)}) \sim \mathcal{N}(\sqrt{\bar{\alpha}_n} X^{(0)}, (1 - \bar{\alpha}_n)I).$$

Since it is non-Markovian process, the reverse process is given by

$$q(X^{(0)}, \dots, X^{(N)}) = q(X^{(N)}) \prod_{n=1}^N q(X^{(n-1)} | X^{(n)}, \dots, X^{(N)}),$$

which we approximate by

$$p_\theta(X^{(0)}, \dots, X^{(N)}) = p_\theta(X^{(N)}) \prod_{n=1}^N p_\theta(X^{(n-1)} | X^{(n)}, \dots, X^{(N)}).$$

Consider the variational lower bound

$$\mathcal{L}(\theta) \stackrel{\text{def}}{=} \mathbb{E}_q \left[\log \frac{p_\theta(X^{(0)}, \dots, X^{(N)})}{q(X^{(1)}, \dots, X^{(N)} | X^{(0)})} \right].$$

Show that

$$\begin{aligned} \mathcal{L}(\theta) &= \mathbb{E}_q [\log p_\theta(X^{(0)} | X^{(1)}, \dots, X^{(N)})] \\ &\quad - \mathbb{E}_q \left[\sum_{n=2}^N D(q(X^{(n-1)} | X^{(0)}) \| p_\theta(X^{(n-1)} | X^{(n)}, \dots, X^{(N)})) \right] \\ &\quad - \mathbb{E}_q [D(q(X^{(N)} | X^{(0)}) \| p_\theta(X^{(N)}))]. \end{aligned}$$

Solution.

$$\begin{aligned} \mathcal{L}(\theta) &= \mathbb{E}_q \left[\log \frac{p_\theta(X^{(0)}, \dots, X^{(N)})}{q(X^{(1)}, \dots, X^{(N)} | X^{(0)})} \right] \\ &= \mathbb{E}_q \left[\log \frac{p_\theta(X^{(N)}) \prod_{n=1}^N p_\theta(X^{(n-1)} | X^{(n)}, \dots, X^{(N)})}{\prod_{n=1}^N q(X^{(n)} | X^{(0)})} \right] \\ &= \mathbb{E}_q [\log p_\theta(X^{(0)} | X^{(1)}, \dots, X^{(N)})] \\ &\quad - \sum_{n=2}^N \mathbb{E}_q \left[\log \frac{q(X^{(n-1)} | X^{(0)})}{p_\theta(X^{(n-1)} | X^{(n)}, \dots, X^{(N)})} \right] \\ &\quad - \mathbb{E}_q \left[\log \frac{q(X^{(N)} | X^{(0)})}{p_\theta(X^{(N)})} \right] \\ &= \mathbb{E}_q [\log p_\theta(X^{(0)} | X^{(1)}, \dots, X^{(N)})] \\ &\quad - \mathbb{E}_q \left[\sum_{n=2}^N D(q(X^{(n-1)} | X^{(0)}) \| p_\theta(X^{(n-1)} | X^{(n)}, \dots, X^{(N)})) \right] \\ &\quad - \mathbb{E}_q [D(q(X^{(N)} | X^{(0)}) \| p_\theta(X^{(N)}))]. \end{aligned}$$